Diskrete Mathematik – Woche 6

Kittitschai Yimvuthikul
https://github.com/tschai-yim/eth-resources
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Relational Mechanics

Problem 1 ⋆ ⋆

Let ρ be a relation on a set A.

1. Prove algebraically that the relation $\rho \circ \hat{\rho}$ is always symmetric.
2. Prove algebraically that if ρ is symmetric, then $\rho \circ \rho$ is symmetric.

Solution:

1.

$$\begin{aligned} \widehat{\rho \circ \hat{\rho}} &= \hat{\hat{\rho}} \circ \hat{\rho} \quad \text{(Lemma 3.8)} \\ &= \rho \circ \hat{\rho} \quad \left(\text{Def. 3.11: } \hat{\hat{\rho}} = \rho\right) \end{aligned}$$

Here $\hat{\hat{\rho}} = \rho$, since $(a,b) \in \hat{\hat{\rho}} \Leftrightarrow (b,a) \in \hat{\rho} \Leftrightarrow (a,b) \in \rho$. Since $\widehat{\rho \circ \hat{\rho}} = \rho \circ \hat{\rho}$, the relation $\rho \circ \hat{\rho}$ is symmetric (Def. 3.15: ρ symmetric $\Leftrightarrow \rho = \hat{\rho}$).

2. Assume $\rho = \hat{\rho}$.

$$\begin{aligned} \widehat{\rho \circ \rho} &= \hat{\rho} \circ \hat{\rho} \quad \text{(Lemma 3.8)} \\ &= \rho \circ \rho \quad \text{(Assumption: } \rho = \hat{\rho}) \end{aligned}$$

Since $\widehat{\rho \circ \rho} = \rho \circ \rho$, the relation $\rho \circ \rho$ is symmetric (Def. 3.15).

Problem 2 ⋆ ⋆ ⋆

Prove that if a relation ρ is symmetric, then its transitive closure ρ^* is also symmetric. (Hint: First prove that $\widehat{\rho^n} = (\hat{\rho})^n$)

Solution:

- **Step 1:** Prove $\widehat{\rho^n} = (\hat{\rho})^n$ by induction.
  - Base case ($n = 1$): $\hat{\rho^1} = \hat{\rho} = (\hat{\rho})^1$.
  - Step: $\widehat{\rho^{k+1}} = \widehat{\rho^k \circ \rho} = \hat{\rho} \circ \widehat{\rho^k}$ (by Lemma 3.7, associativity, and Lemma 3.8).
  - By IH, $\hat{\rho} \circ (\hat{\rho})^k = (\hat{\rho})^{k+1}$ (Lemma 3.7, associativity).
- **Step 2:** Show ρ^n is symmetric.
  - Because ρ is symmetric, $\hat{\rho} = \rho$.
  - Therefore, $\widehat{\rho^n} = (\hat{\rho})^n = \rho^n$. Thus, every power ρ^n is symmetric.
- **Step 3:** Show ρ^* is symmetric.
  - $(a,b) \in \rho^* \Rightarrow (a,b) \in \rho^k$ for some $k \geq 1$.
  - Since ρ^k is symmetric, $(b,a) \in \rho^k$.
  - Since $\rho^k \subseteq \rho^*$, it follows that $(b,a) \in \rho^*$.

Relation Properties

Problem 3 ⋆ ⋆

Let ρ and σ be relations on the set $\mathbb{Z}$.

1. If ρ and σ are transitive, is $\rho \cup \sigma$ always transitive? Prove or provide a minimal counterexample.
2. If ρ and σ are symmetric, is $\rho \cup \sigma$ always symmetric? Prove or provide a minimal counterexample.

Solution:

1. **False.** Let $\rho = \{(1,2)\}$ and $\sigma = \{(2,1)\}$. Both are (vacuously) transitive.
   - $\rho \cup \sigma = \{(1,2),(2,1)\}$.
   - This union is not transitive because $(1,2),(2,1) \in \rho \cup \sigma$ but $(1,1) \notin \rho \cup \sigma$.
   - This is minimal: a relation with a single pair is always transitive, so the union needs at least two pairs, and with only one element of $\mathbb{Z}$ every relation is transitive.
2. **True.** Assume $(a,b) \in \rho \cup \sigma$.
   - Case 1: $(a,b) \in \rho$. Since ρ is symmetric, $(b,a) \in \rho \Rightarrow (b,a) \in \rho \cup \sigma$.
   - Case 2: $(a,b) \in \sigma$. Since σ is symmetric, $(b,a) \in \sigma \Rightarrow (b,a) \in \rho \cup \sigma$.
   - In all cases, $(b,a) \in \rho \cup \sigma$.

Problem 4 ⋆ ⋆

Let ρ be a relation on A.

1. Disprove the claim: "If ρ is irreflexive, then ρ^2 is irreflexive."
2. Prove the claim: "If ρ is irreflexive and transitive, then ρ is antisymmetric."

Solution:

1. Let $A = \{1,2\}$ and $\rho = \{(1,2),(2,1)\}$.
   - ρ is irreflexive because $(1,1) \notin \rho$ and $(2,2) \notin \rho$.
   - $\rho^2 = \{(1,1),(2,2)\}$.
   - ρ^2 is not irreflexive; in fact, $\rho^2 = \mathrm{id}_A$, which is reflexive.
2. Assume ρ is irreflexive and transitive, and assume $a\rho b$ and $b\rho a$.
   - By transitivity, $a\rho a$.
   - This contradicts the irreflexivity of ρ.
   - So the premise of antisymmetry can never hold, and ρ is (vacuously) antisymmetric.

Problem 5 ⋆ ⋆ (extra)

Let ρ and σ be antisymmetric relations on a set A.

1. Prove that the intersection $\rho \cap \sigma$ is antisymmetric.
2. Disprove that the composition $\rho \circ \sigma$ is always antisymmetric.

Solution:

1. Assume $(a, b) \in \rho \cap \sigma$ and $(b, a) \in \rho \cap \sigma$.
   - This means $(a, b) \in \rho \wedge (b, a) \in \rho$, and $(a, b) \in \sigma \wedge (b, a) \in \sigma$.
   - Since ρ is antisymmetric, $(a, b) \in \rho \wedge (b, a) \in \rho \Rightarrow a = b$.
   - Thus, $\rho \cap \sigma$ is antisymmetric. (The σ properties were not even needed).
2. **False.** Let $A = \{1, 2, 3, 4\}$, $\rho = \{(1, 2), (3, 4)\}$ and $\sigma = \{(2, 3), (4, 1)\}$.
   - Both ρ and σ are antisymmetric (they have no symmetric pairs).
   - $\rho \circ \sigma = \{(1, 3), (3, 1)\}$.
   - $\rho \circ \sigma$ contains $(1, 3)$ and $(3, 1)$ but $1 \neq 3$, so it is not antisymmetric.

Equivalence Relations

Problem 6 ⋆ ⋆

Let $f : A \to B$ be a function. Define a relation on A by $x \underset{f}{\sim} y \Leftrightarrow f(x) = f(y)$.

1. Briefly show this is an equivalence relation.
2. Let $f : \mathbb{Z} \to \mathbb{Z}$ be $f(x) = |x| \bmod 3$. Explicitly list the equivalence classes of the quotient set $\frac{\mathbb{Z}}{\underset{f}{\sim}}$.

Solution:

1. - Reflexive: $f(x) = f(x) \Rightarrow x \underset{f}{\sim} x$.
   - Symmetric: $f(x) = f(y) \Rightarrow f(y) = f(x) \Rightarrow y \underset{f}{\sim} x$.
   - Transitive: $f(x) = f(y) \wedge f(y) = f(z) \Rightarrow f(x) = f(z) \Rightarrow x \underset{f}{\sim} z$.
2. The function maps any integer to the remainder of its absolute value divided by 3. The possible outputs are 0, 1, and 2.
   - $[0] = \{x \in \mathbb{Z} \mid |x| \bmod 3 = 0\} = \{3k \mid k \in \mathbb{Z}\}$
   - $[1] = \{x \in \mathbb{Z} \mid |x| \bmod 3 = 1\} = \{3k + 1 \mid k \geq 0\} \cup \{3k - 1 \mid k \leq 0\}$
   - $[2] = \{x \in \mathbb{Z} \mid |x| \bmod 3 = 2\} = \{3k + 2 \mid k \geq 0\} \cup \{3k - 2 \mid k \leq 0\}$

   The quotient set is $\frac{\mathbb{Z}}{\underset{f}{\sim}} = \{[0], [1], [2]\}$.

Problem 7 ⋆ ⋆ ⋆

Define a relation ρ to be **circular** if $\forall a, b, c : (a \rho b \wedge b \rho c) \Rightarrow c \rho a$. Prove the following theorem: A relation ρ is an equivalence relation if and only if it is both reflexive and circular.

Solution:

($\Rightarrow$) If ρ is an equivalence relation, it is reflexive, symmetric, and transitive.
- Assume $a\rho b \wedge b\rho c$.
- By transitivity, $a\rho c$.
- By symmetry, $c\rho a$. Thus, it is circular.

($\Leftarrow$) Assume ρ is reflexive and circular.
- **Symmetry**: Assume $a\rho b$. Because ρ is reflexive, we know $a\rho a$.
  - Using circularity on (a, a, b): $a\rho a \wedge a\rho b \Rightarrow b\rho a$. Thus, ρ is symmetric.
- **Transitivity**: Assume $a\rho b \wedge b\rho c$.
  - By circularity, $c\rho a$.
  - Since we just proved ρ is symmetric, $c\rho a \Rightarrow a\rho c$. Thus, ρ is transitive.

Problem 8 (extra)

Given an arbitrary relation ρ on A. Let $\sigma_1 = \rho \cup \hat{\rho} \cup \mathrm{id}_A$.

1. ($\star$) Is σ_1 always an equivalence relation? Justify your answer.
2. ($\star\star\star$) Express the **smallest** equivalence relation σ_2 containing ρ using union, inverse, identity and transitive closure, and prove that your expression is correct.

Solution:

1. **No.** σ_1 is guaranteed to be reflexive (due to $\cup\, \mathrm{id}_A$) and symmetric (due to $\cup\, \hat{\rho}$). However, it is not necessarily transitive.
   - E.g. $A = \{1, 2, 3\}$ and $\rho = \{(1, 2), (2, 3)\}$: σ_1 contains $(1, 2)$ and $(2, 3)$, but not $(1, 3)$.
2. The transitive closure has to be applied after the symmetrisation (e.g. $\rho^* \cup \hat{\rho}^* \cup \mathrm{id}_A$ is not transitive in general).
   - $\sigma_2 = (\rho \cup \hat{\rho} \cup \mathrm{id}_A)^*$
   - Proof: Let $\tau = \rho \cup \hat{\rho} \cup \mathrm{id}_A$, so $\sigma_2 = \tau^*$.
     1. $\tau^* \supseteq \tau \supseteq \mathrm{id}_A$, so τ^* is reflexive.
     2. τ is symmetric, so τ^* is symmetric by Problem 2.
     3. τ^* is transitive: if $a\tau^m b$ and $b\tau^n c$, then $a\tau^{m+n} c$.
     4. Minimality: let θ be any equivalence relation with $\rho \subseteq \theta$. Then $\mathrm{id}_A \subseteq \theta$ by reflexivity and $\hat{\rho} \subseteq \hat{\theta} = \theta$ by symmetry, so $\tau \subseteq \theta$. By transitivity ($\theta^n \subseteq \theta$), $\tau^n \subseteq \theta^n \subseteq \theta$ for all $n \geq 1$, hence $\tau^* \subseteq \theta$.

Partial Orders

Problem 9 $\star$

We know $(\mathbb{N}; \leq)$ is well-ordered.

1. Is $(\mathbb{Z}; \leq)$ well-ordered? Justify your answer.
2. Is the subset of rational numbers $S = \{q \in \mathbb{Q} \mid 0 \leq q \leq 1\}$ well-ordered by $\leq$? Justify your answer.

Solution:

A poset is well-ordered if it is totally ordered and every non-empty subset has a least element (Def. 3.30). Both $\mathbb{Z}$ and S are totally ordered by $\leq$, so only the second condition is in question.

1. **No.** Consider the subset $\mathbb{Z}$ itself (or the negative integers $\{-1, -2, -3, \ldots\}$). This non-empty subset extends infinitely downwards and has no least element.
2. **No.** Consider the subset $S' = \{q \in \mathbb{Q} \mid 0 < q \leq 1\}$. S' is non-empty. For any rational $q > 0$ in S', the rational number $\frac{q}{2}$ is strictly smaller and also in S'. Therefore, S' has no least element.

Problem 10 ⋆ ⋆

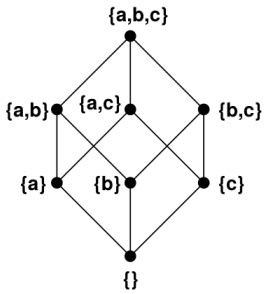
In any partially ordered set $(A; \preceq)$, let $S \subseteq A$. Formally prove that if S has a least upper bound (supremum), it is unique.

Solution:

- Assume that x and y are both least upper bounds of S.
- Since x is a **least** upper bound, and y is an upper bound, it must be that $x \preceq y$.
- Since y is a **least** upper bound, and x is an upper bound, it must be that $y \preceq x$.
- By the antisymmetry property of the partial order $\preceq$, $x \preceq y \land y \preceq x \Rightarrow x = y$.
- Therefore, the least upper bound is unique.

Problem 11 ⋆ (extra)

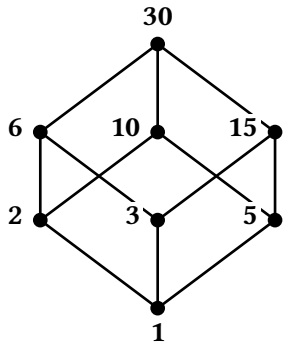
Consider the Boolean lattice $(\mathcal{P}(\{a, b, c\}); \subseteq)$, which forms a cube in its Hasse diagram (see Figure 3.1 in script). Find a set of 8 integers X such that the poset $(X; |)$ has the exact same Hasse diagram structure.



Solution:

To map the power set lattice to divisibility, we use the prime factorization.

- Assign each element of the base set to a distinct prime number: $\varphi : a \mapsto 2, b \mapsto 3, c \mapsto 5$.
- Map subsets to the product of their primes (the empty set maps to the empty product, 1).
- $X = \{1, 2, 3, 5, 6, 10, 15, 30\}$.
- Subset inclusion perfectly mirrors divisibility here: $A \subseteq B \Leftrightarrow \prod_{s \in A} \varphi(s) \mid \prod_{s \in B} \varphi(s)$, with the empty product $= 1$.
- Hasse diagram of $(X; |)$ (12 cover relations: 1–2, 1–3, 1–5, 2–6, 2–10, 3–6, 3–15, 5–10, 5–15, 6–30, 10–30, 15–30):



Problem 12 ⋆ ⋆ (extra)

Let $\left(A; \preceq_A\right)$ and $\left(B; \preceq_B\right)$ be two posets. Recall the direct product relation $\leq$ on $A \times B$ from Definition 3.28: $(a_1, b_1) \leq (a_2, b_2) \Leftrightarrow a_1 \preceq_A a_2 \wedge b_1 \preceq_B b_2$.

Formally prove that $\leq$ is antisymmetric on $A \times B$.

Solution:

- Assume $(a_1, b_1) \leq (a_2, b_2)$ and $(a_2, b_2) \leq (a_1, b_1)$.
- By definition of $\leq$, the first statement means: $a_1 \preceq_A a_2 \wedge b_1 \preceq_B b_2$.
- The second statement means: $a_2 \preceq_A a_1 \wedge b_2 \preceq_B b_1$.
- Grouping the conditions for A: we have $a_1 \preceq_A a_2 \wedge a_2 \preceq_A a_1$. Since $\preceq_A$ is a partial order, it is antisymmetric, so $a_1 = a_2$.
- Grouping the conditions for B: we have $b_1 \preceq_B b_2 \wedge b_2 \preceq_B b_1$. Since $\preceq_B$ is antisymmetric, $b_1 = b_2$.
- Therefore, $(a_1, b_1) = (a_2, b_2)$. Thus, $\leq$ is antisymmetric.

Problem 13 ⋆ ⋆ ⋆ (extra)

Consider the claim: "If a partially ordered set $(A; \preceq)$ has exactly one minimal element m, then m is the least element of A."

1. Prove this claim is true if A is **finite**. (Hint: use a proof by contradiction).
2. Disprove this claim for **infinite** posets by describing a counterexample.

Solution:

1. Assume A is finite, has a unique minimal element m, but m is **not** the least element.
   - This means $\exists x \in A$ such that $\neg(m \preceq x)$.
   - Start a sequence at $x_0 = x$. Since x_0 is not minimal (otherwise it would equal m, and $m \preceq m$ contradicts $\neg(m \preceq x)$), there exists $x_1 \prec x_0$.
   - We can continue this strictly decreasing chain $x_k \prec \ldots \prec x_1 \prec x_0$: every x_k satisfies $x_k \preceq x_0 = x$ by transitivity, so it is not minimal either (if x_k were minimal, $x_k = m$, and $m = x_k \preceq x_0 = x$ would contradict $\neg(m \preceq x)$). Hence some $x_{k+1} \prec x_k$ always exists.
   - The x_i are pairwise distinct, since $x_i \prec x_j$ for $i > j$ by transitivity of $\preceq$ and antisymmetry. So the chain has at most $|A|$ elements.
   - This contradicts the fact that the chain can always be extended. Hence m is the least element.
2. Consider the infinite set $A = \{a, b_1, b_2, b_3, \ldots\}$.
   - Define $b_j \preceq b_i :\Leftrightarrow j \geq i$, $a \preceq a$, and no other relations. This is a partial order: it is $\geq$ on the indices plus an isolated point a. In particular $b_{i+1} \prec b_i$ for all i, and a is incomparable to all b_i.
   - The sequence of b_i goes down infinitely, so no b_i is minimal.
   - The element a has nothing smaller than it, so a is minimal. It is the **unique** minimal element.
   - However, a is not the **least** element because $\neg(a \preceq b_1)$.